

Zihao Zhao

zihaoz@illinois.edu | neo1zh.github.io | Google Scholar | LinkedIn

RESEARCH INTERESTS

Reasoning, learning, and efficient orchestration in LLM-based multi-agent systems.

EDUCATION

University of Illinois Urbana-Champaign Sept. 2026–Present

Ph.D. student in Information Sciences Advisor: Hao Wang

Rutgers University–New Brunswick Aug. 2025–Aug. 2026

Doctoral studies in Computer Science Advisor: Hao Wang

Shanghai Jiao Tong University Sept. 2021–June 2025

B.Eng. in Automation, Zhiyuan Honors Program (top 5% of cohort)

PUBLICATIONS

Published and accepted papers. † Equal contribution.

1. **Zihao Zhao**, Tunyu Zhang, Haizhou Shi, Yusong Zhao, Xinxi Zhang, and Hao Wang
When Debate Helps: Proposal Supply and Verification-Aware Readout in Multi-Agent Reasoning
NeurIPS, 2026. Accepted. [paper]
2. King Yeung Tsang†, **Zihao Zhao**†, Vishal Venkataramani, Haizhou Shi, Zixuan Ke, Semih Yavuz, Shafiq Joty, and Hao Wang
Reward Modeling for Multi-Agent Orchestration
NeurIPS, 2026. Accepted. [paper]
3. Tunyu Zhang, **Zihao Zhao**, Yusong Zhao, Haizhou Shi, Zhuohang Li, Haoxian Chen, Hao Wang, and Dimitris N. Metaxas
Sequential Probabilistic Uncertainty Estimation for Parallel Multi-Agent Reasoning Systems
NeurIPS, 2026. Accepted. [paper]
4. Wenyuan Wang, Yusong Zhao, ..., **Zihao Zhao**, ..., Hao Wang
Probabilistic Residual Learning for Online Recommendations
RecSys, 2026. Accepted. [paper]
5. Yiyang Zhou, Linjie Li, ..., **Zihao Zhao**, ..., Huaxiu Yao
GLIMPSE: Do Large Vision-Language Models Truly Think With Videos or Just Glimpse at Them?
EMNLP, 2025, pp. 27842–27856. [paper]
6. Sihan Yang, Chenhang Cui, **Zihao Zhao**, Yiyang Zhou, Weilong Yan, Ying Wei, and Huaxiu Yao
Improving Alignment in LVLMs with Debaised Self-Judgment
Findings of EMNLP, 2025. [paper]
7. Nishchal Sapkota, Yejia Zhang, **Zihao Zhao**, Maria Jose Gomez, Yuhan Hsi, Jordan A. Wilson, Kazuhiko Kawasaki, Greg Holmes, Meng Wu, Ethylin Wang Jabs, Joan T. Richtsmeier, Susan M. Motch Perrine, and Danny Z. Chen
Universal Conditional Networks (UniCoN) for Multi-Age Embryonic Cartilage Segmentation with Sparsely Annotated Data
Scientific Reports, 2025, vol. 15, article 3893. [paper]

8. Yunfei Luan, Muhang He, Yudong Tian, Chengjie Lin, Yunhan Fang, **Zihao Zhao**, Jianxin Yang, and Yao Guo
Intelligent Disinfection Robot with High-Touch Surface Detection and Dynamic Pedestrian Avoidance
ICRA, 2024, pp. 3595–3601. [paper]

MANUSCRIPTS UNDER REVIEW

9. Yusong Zhao, **Zihao Zhao**, Zirui Zhang, Tunyu Zhang, Ziyan Wang, Chengzhi Mao, Dimitris N. Metaxas, and Hao Wang
SAEPO: Sparse-Autoencoder-Guided Token-Level Credit Assignment for Policy Optimization
Manuscript under review, 2026.

TEACHING EXPERIENCE

University of Illinois Urbana-Champaign

Teaching Assistant: IS 206 Database.

Rutgers University–New Brunswick

Teaching Assistant: CS 211 Computer Architecture; CS 462 Introduction to Deep Learning.

PROFESSIONAL EXPERIENCE

SJTU Network & Information Management Office

Senior Student Network Administrator, Core Team

Sept. 2021–Sept. 2023

Shanghai, China

- Maintained campus wireless infrastructure and resolved connectivity issues for 40,000 students and faculty members.
- Authored technical documentation and mentored two junior administrators; contributed more than 300 hours of service.

AWARDS & HONORS

- **National First Prize**, TI Cup China Undergraduate Electronics Design Contest 2023
Top 1%; one of only two SJTU teams to receive the National First Prize.
- **Enterprise Scholarship**, Shanghai Jiao Tong University 2023
CNY 10,000; one of the top two students in the Zhiyuan Honors Program.
- **Zhiyuan Honor Scholarship**, Shanghai Jiao Tong University (CNY 15,000) 2021–2023
- **President’s Prize**, Shanghai Jiao Tong University 2022
- **Merit Student**, Shanghai Jiao Tong University 2022
- **Alumni Scholarship**, Shanghai Jiao Tong University 2022
- **First Prize**, Chinese Chemistry Olympiad (top 1%) 2020

ACADEMIC SERVICE

Reviewer: ACL Rolling Review (2025–2026); NeurIPS (2026).